

No Free Credit: How Much Simulated Evidence Can Replace Real Events in Zero-Failure Reliability Demonstration of Perception Systems

Lam Quoc Dat¹

School of Business and Technology (FSB), FPT University, Ho Chi Minh City, Vietnam

Abstract

A perception system on a new fixed camera is often accepted by a zero-failure reliability demonstration: a fixed number of real events, the zero-failure sample size (59 events for 95% confidence that at most 5% of events are missed), must pass without a miss. How many can simulated evidence replace? We prove that, without an assumption linking the simulator to reality, independent simulated data cannot shorten the demonstration; that accepting a larger error probability buys an exactly computable number of events; and that a synthetic twin paired with each real event bounds the real miss rate by the twin miss rate plus a discordance rate, which pays off only through a transfer bound priced in calibration scenes. On 76 fixed-camera scenes, real misses are rare at the pre-specified operating point (0.6%), so it is untested there. If calibration carries over, a twin run about 23 times per camera would save at most 14 events per camera at a 20% target; the twin as built saves none. The transfer certificate stands at scene level but is withdrawn at site level (21 sites). Two named failure modes: night and camera-jitter cameras pass through (4 of 6 and 1 of 2 falsely accepted).

Keywords: reliability demonstration, zero-failure testing, synthetic data, digital twin, success-run, perception system, transfer bound

Email address: lamquocdat@gmail.com (Lam Quoc Dat)

¹ORCID: 0009-0004-5432-9343

1. Introduction

A reliability demonstration test (RDT) accepts a product when it passes a planned test; for attribute (pass/fail) data the classical plan is the zero-failure, or success-run, test [1, 2]. If each demand fails independently with probability p , a unit with $p > \varepsilon$ passes n demands with probability at most $(1-\varepsilon)^n$, so passing $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1-\varepsilon) \rceil$ demands without failure demonstrates $p \leq \varepsilon$ at confidence $1 - \delta$: 59 demands at $\varepsilon = \delta = 5\%$. Zero-failure planning remains common industrial practice [3], and its sample-size burden is a recurring motive for test-plan research in this journal, from success-failure verification tests [4] to qualification plans that borrow auxiliary information [5, 6].

We consider a learned perception component, a fixed object detector on a fixed camera that samples frames periodically, as the unit under test. A demand is the passage of an object (an *event*); a failure is an event on which no sampled frame yields a detection. Such components are increasingly assessed with reliability-engineering tools [7, 8, 9]. For them, the demands are rare and every new installation is a new unit: 59 failure-free events must be collected per camera, and for road vehicles the exposure needed to demonstrate rare-failure reliability by field operation alone is prohibitive [10].

Simulation is the obvious remedy. Digital twins and simulators already supply reliability evidence in engineering practice [11, 12, 13], and virtual scenario testing is a leading proposed alternative to mileage-based demonstration of automated vehicles [14, 15]. For perception, however, the simulation-to-reality gap is measurable and has been measured under different environmental conditions [16, 17], and performance in simulation need not predict performance in the field [18]. The question for a demonstration test is therefore quantitative: *how many of the n_0 real demands can simulated evidence replace, and what form of guarantee survives when the evidence is transferred to an untested unit?*

This paper answers the question for zero-failure demonstration of a perception component.

1. *No free credit.* Without an assumption that links the simulator to reality, no valid procedure can use independent simulated data to shorten a zero-failure demonstration, for any simulator and any amount of simulation. This makes precise, for success-run tests, a principle known from clinical borrowing [19], and relates to the limits on combining evidence in software dependability [20].

- 38 2. *The exact price of relaxing confidence.* Accepting error probability $\delta' >$
39 δ when the simulator is wrong buys at most, and exactly, $n_0(\varepsilon, \delta) -$
40 $n_0(\varepsilon, \delta')$ demands: 8, 14, 22 and 27 of the 59 for $\delta' = 7.5, 10, 15$ and
41 20%.
- 42 3. *A paired-twin bound and a fleet-level transfer bound.* A synthetic twin
43 of each real event bounds the real failure probability by the twin failure
44 probability plus a discordance rate, with no assumption; on one camera
45 this costs more real events than the plain test (72–122). Calibrated
46 once across scenes, the discordance transfers to an untested camera as
47 a population-average bound whose price is the number of calibration
48 scenes, not the number of events.
- 49 4. *An exchange rate on public benchmarks, and a named failure mode.*
50 On 76 fixed-camera scenes the twin as built saves no real event; a twin
51 run to convergence would save at most 14 real events per camera, if
52 the discordance calibrated with paired twins also held for the twins of
53 new cameras (not verified), at the operating point fixed in advance,
54 where real failures are rare (0.61%) and the demonstration is therefore
55 untested. The transfer certificate stands at scene level but is withdrawn
56 when the unit is the site (21 sites). When out-of-domain cameras are
57 judged alone, they pass through: 4 of 6 night cameras and, at a fixed
58 sampling phase, 1 of 2 camera-jitter cameras are falsely accepted; we
59 call this *out-of-domain pass-through*, with two named modes, night and
60 camera jitter.

61 *Reliability demonstration tests in the zero-failure regime have no estab-*
62 *lished rule for how much simulated evidence may substitute for real evidence,*
63 *nor a guarantee form that survives transfer to an untested unit; this paper*
64 *gives a finite-sample answer (none without an assumption; an exact price*
65 *for relaxing confidence; a fleet-level bound whose price is scene count) and*
66 *measures it on 76 fixed-camera scenes.*

67 2. Related work and research gap

68 Table 1 positions the paper against four bodies of work.

69 (a) *Reliability demonstration and zero-failure test planning.* The zero-
70 numerator bound and the rule of three are classical [1, 2], and the Bayesian
71 zero-failure (BAZE) test sets the required test time through a prior on the

Table 1: Research gap. Unit: what the guarantee covers. ZF: is the zero-failure (success-run) regime treated? Price: is the real-evidence value of simulation quantified? Transfer: does the guarantee reach an untested unit? Link: assumption needed between simulation and reality. Evidence: empirical basis.

Body of work	Unit	ZF	Price of simulation	Transfer	Sim-real link	Evidence
(a) Reliability demonstration, zero-failure planning [1, 21, 22, 23, 4, 5, 24]	product / unit	yes	through a prior or auxiliary data; not as real events saved	to the tested product; prior-system evidence handled conservatively	prior or similarity assumption (Bayesian); none (frequentist, no simulation)	case studies
(b) Simulation-assisted testing, digital-twin reliability [11, 12, 13, 25, 26]	system	no	fidelity weighting in fusion	not addressed (extrapolation named as open)	fidelity or calibration model	engineering case studies
(c) Simulation-augmented evaluation of perception [16, 17, 27, 15, 28, 29]	mean metric	no	gap measured; variance reduction for means	no	paired labels (rectifier) or none	perception benchmarks
(d) Finite-sample guarantees and assurance for learned perception [30, 31, 32, 33, 8]	per deployment	no	not addressed (real calibration data)	exchangeability of calibration and deployment	exchangeability	detection data sets
(e) This paper	fleet (population average); per-unit as a data requirement	yes	exact: none without a link; $n_0(\delta) - n_0(\delta')$ when relaxing; 14 events per camera at most	yes, priced by scene count $(U(J, m) = 3.87\%, J = 0, m = 76)$	none for the twin bound; scenes exchangeable (i.i.d.) for transfer, as in (d) at scene level	76 fixed-camera scenes, 164 events; failure mode measured

72 failure rate [21]. Recent work designs sample sizes by assurance [22], stud-
73 ies the limits of Bayesian zero-failure plans [3], and publishes in this jour-
74 nal demonstration and acceptance plans for degrading products [23, 34],
75 success-failure verification tests [4] and qualification plans that use subsys-
76 tem, Monte-Carlo or expert information [5, 6]. For software and autonomous
77 systems, conservative Bayesian inference uses prior knowledge, including ev-
78 idence from previous similar systems, without optimistic bias [24, 35, 36],
79 and reusing the reliability of an older system as a prior is optimistic rather
80 than conservative [37]. Bayesian borrowing quantifies how much a prior is
81 worth through its effective sample size [38] and discounts historical data

through power or commensurate priors [39, 40], and conservative frequentist bounds have been derived when the test profile differs from the operational profile [41]. *What is missing:* a frequentist statement of how much the zero-failure sample size can be reduced by simulated rather than prior-system evidence, and at what price.

(b) *Simulation-assisted testing and digital-twin reliability.* Digital twins feed reliability models with simulated operating data [11], are validated against physical experiments [12], and are discussed as maintenance tools whose suitability is still open [42]. Multi-fidelity fusion combines scarce physical tests with abundant simulation for mission-reliability prediction, weighting the simulation by its fidelity gap [13], and simulators generate test scenarios for autonomous systems [25, 14]. Model verification, validation and uncertainty quantification identify extrapolation to untested conditions as a major challenge [26]. *What is missing:* a rule that states which assumption about the simulator is needed before its output may reduce a zero-failure test, and a measured sim–real discordance that makes the assumption checkable.

(c) *Simulation-augmented evaluation of perception.* The simulation-to-reality gap of camera object detectors has been measured across day, night, fog and rain [16] and for drone detection [17]; synthetic images are used to test driving perception, with a distribution mismatch that biases test results [27], and perception digital twins reproduce field data only approximately [43]. Statistical model checking evaluates perception in simulation only [44], simulation performance can fail to predict field performance [18], and a statistical foundation for aligning synthetic and real testing outcomes has been called for [15]. Closest to our question, Sim2Val uses simulator outputs as control variates to reduce the number of real samples needed for a confidence bound on mean performance (CoRL 2025, arXiv preprint) [45], SCAPE combines limited real rollouts with simulation proxies for policy evaluation (preprint) [46], and multi-fidelity fusion weights simulation by its fidelity gap in reliability prediction [13]; none treats the zero-failure regime. Prediction-powered inference and its stratified variant combine many proxy outcomes with few real labels through a paired rectifier [28, 29]. *What is missing:* these methods estimate a mean performance or reduce its variance; none addresses the zero-failure regime, where the quantity to bound is a rare failure probability and the rectifier itself would cost almost n_0 real events per unit.

(d) *Finite-sample guarantees and assurance for learned perception.* Conformal prediction gives distribution-free finite-sample guarantees [47], cali-

brated risk control extends them to general losses [30], and they have been applied to object detection [31], keypoint-based pose estimation [48], perception for safe navigation [32] and, in this journal, anomaly detection in cyber-physical systems [33]. Reliability of machine-learning components is also modelled directly, through multi-version architectures [7], safety-assurance cases for object detectors [8] and failure-free-operation definitions under an operational profile [9]. *What is missing:* these guarantees are calibrated on real data from the deployment distribution; they do not say how much of that calibration data may be synthetic, nor how a guarantee transfers to a unit whose own data are absent.

3. Demonstration setting

Unit, demand and failure. The unit under test is a fixed camera s with a fixed detector. The camera observes *events*: maximal runs of frames that contain the target object. It samples frames with period K and an unknown phase φ , and runs the detector on the sampled frames. An event is *missed* ($M = 1$) if no sampled frame is a hit; with H the hit frames of an event and φ uniform, $\mathbb{P}(M = 1) = 1 - |H \bmod K|/K$. The miss probability of camera s is p_s , and the demonstration goal is the assurance claim $p_s \leq \varepsilon$ at confidence $1 - \delta$. $U(x, n)$ denotes the one-sided $(1 - \delta)$ Clopper–Pearson upper bound for x failures in n demands. We call an accepted assurance claim of this form a *certificate*. In the language of acceptance sampling, δ is the consumer’s risk; the producer’s risk is reported as the power of a plan at the observed miss rate. All claims concern the miss probability averaged over the uniform sampling phase; a camera deployed with one fixed, unknown phase is covered only on average over phases.

Simulated evidence and twins. A simulator produces data $Y \sim Q$. A *paired twin* renders a synthetic counterpart of each real event and runs the same detector at the same sampling phase, giving M' . We write $q = \mathbb{P}(M' = 1)$ and $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch), the *discordance*. Every rate in this paper is per event; with three sprite variants per event, the twin catches an event if at least two variants do, and an event counts as discordant if it is discordant at one or more sampling phases (worst phase; an integer count, conservative with respect to the phase-averaged rate, which we give in parentheses). Table 2 collects the notation.

Table 2: Notation.

symbol	meaning
ε, δ	target miss probability; allowed probability of a false acceptance
$n_0(\varepsilon, \delta)$	zero-failure sample size $\lceil \ln \delta / \ln(1 - \varepsilon) \rceil$
$K, \varphi, d_{\min}$	sampling period, sampling phase, minimum event length (frames)
M, M'	real miss, twin miss of an event at the same phase
p_s, q_s, D_s	real miss probability, twin miss probability, discordance $\mathbb{P}(M = 1, M' = 0)$ of camera s
m, n_s, N	calibration scenes, paired events of scene s , $N = \sum_s n_s$
X, J	discordant events; scenes with at least one discordant event
$U(x, n)$	one-sided $(1 - \delta)$ Clopper–Pearson upper bound
fleet bound, population bound	$U(X, N)$ and $U(J, m)$ (Result 4)
OP*, OP-A, OP-C	operating points $(\varepsilon, d_{\min}, K) = (20\%, 32, 16)$, $(20\%, 32, 8)$, $(10\%, 32, 1)$

154 *Random model.* Scenes are drawn from a population; given a scene, events
155 are drawn i.i.d. from the scene’s event law and each receives an independent
156 uniform phase; given the event and the phase, the real and twin outcomes
157 are determined (fixed detector, fixed sprite seeds). Every per-event indicator
158 below—real miss, twin miss, discordance at a random phase, and the worst-
159 phase indicator—is therefore a Bernoulli variable, independent across events
160 given the scene. In the audit, the “truth” of a scene is its empirical event
161 law, resampled with replacement.

162 4. What simulated evidence can buy: four results

163 The four results below are stated in words; Appendix A gives the formal
164 statements, the assumptions and the proofs, each with a numerical check.

165 *Result 1 (no free credit).* Suppose the simulated data are independent of the
166 real demands and no assumption links the simulator to reality, so that for
167 every simulator law Q both a perfect unit ($p = 0$) and a unit just worse
168 than the target ($p \downarrow \varepsilon$) are possible. Then any acceptance procedure that
169 is valid at confidence $1 - \delta$ accepts a perfect unit with probability at most

Table 3: Result 2: exact credit c versus the continuous form c^* ($\varepsilon = \delta = 5\%$, $n_0 = 59$), and the two-tier plan run with the twin event pool as an independent simulator (worst case $p = \varepsilon$; closed form given the pass probability, and 100,000 Monte-Carlo runs).

δ'	7.5%	10%	15%	20%
exact credit c (demands)	8	14	22	27
saving w.r.t. n_0	14%	24%	37%	46%
continuous form c^*	7	13	21	27
false acceptance, twin pool (closed form)	7.3%	9.8%	14.8%	19.1%
Monte Carlo	7.2%	9.7%	14.7%	19.1%

170 $\delta/(1 - \varepsilon)^n$ when it uses n real demands, whatever the simulator and however
171 much simulation it uses. Accepting a perfect unit with certainty therefore
172 still needs $n \geq n_0(\varepsilon, \delta)$ real demands. The reason is that the simulated
173 data have the same law under both hypotheses, while an all-success real
174 sample is possible under both. This makes precise, for the success-run test,
175 the principle that external evidence cannot shorten a test whose validity
176 must hold for every external law [19], and the related limit on combining
177 evidence for ultra-high dependability [20]. A paired twin is built from the
178 real events and is therefore *not* an independent simulator; pairing is what
179 Result 3 exploits.

180 *Result 2 (the exact price of relaxing confidence).* If the demonstration may
181 accept a worse unit with probability $\delta' > \delta$ when the simulator is wrong—
182 that is, if one accepts a higher consumer’s risk—the real demands can be
183 reduced by at most

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta'),$$

184 and this reduction is attained by a two-tier plan: run the zero-failure test with
185 $n_0(\varepsilon, \delta')$ demands when a simulator test passes and with $n_0(\varepsilon, \delta)$ otherwise.
186 The plan keeps confidence $1 - \delta$ only if a simulator of a worse-than-target
187 unit passes with probability at most π^* : 6.1%, 3.0%, 1.5% and 1.0% for
188 $\delta' = 7.5, 10, 15$ and 20% at $\varepsilon = \delta = 5\%$. This requirement is itself a
189 link between simulator and reality, so Result 2 is best read as the price of
190 lowering the confidence, not as a free credit. The frequently used continuous
191 form $\lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$ is only a lower bound on c . Table 3 gives the
192 numbers.

193 Monte-Carlo runs match the closed form within 0.29 percentage points on

194 a 40-point grid. Used as an independent simulator (its events resampled in-
 195 dependently of the real draws), the twin event pool passes the simulator test
 196 in 98.1% of runs, the credit is almost fully realised, and the false-acceptance
 197 probability at $p = \varepsilon$ is δ' up to the pass probability: the credit is bought ex-
 198 actly with the extra error. An off-the-shelf synthetic benchmark (BMC [49];
 199 30 events long enough for the operating point) never passes at $\varepsilon = 5\%$ (0%):
 200 it has too few synthetic events to bound its own miss probability below 5%.
 201 At $\varepsilon = 20\%$ it passes in every run and buys the same credit as the twin
 202 (supplementary Table S13), which shows that passing the simulator test is
 203 no evidence of fidelity. Monte-Carlo standard errors are about 0.1%.

204 *Result 3 (paired-twin bound).* For any pairing of real and twin events, the real
 205 miss probability is at most the twin miss probability plus the discordance,

$$p \leq q + D,$$

206 with no assumption on the simulator (the bound splits the real misses into
 207 those the twin also misses and those it catches). On a single camera the
 208 bound does not save demands: if $q \leq q_U$ holds with confidence $1 - \delta/2$ and
 209 $D \leq \varepsilon - q_U$ is demonstrated by a zero-discordance run at level $\delta/2$, the
 210 camera needs $\lceil \ln(\delta/2)/\ln(1 - \varepsilon + q_U) \rceil$ paired events, i.e. 72, 91 and 122
 211 for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all above $n_0 = 59$. A twin built with a
 212 pessimistic (shorter) event length remains valid: its miss probability can only
 213 grow and its discordance only shrink. On the fixed-camera twin, shortening
 214 every event by 50% raises the twin miss rate at OP* from 0.95% to 1.68%.
 215 With a uniform phase the discordance probability of an event is $|R' \setminus R|/K$,
 216 where R, R' are the residues modulo K of the real and twin hit frames; the
 217 worst-phase indicator $\mathbf{1}\{R' \setminus R \neq \emptyset\}$ dominates it, so the bounds of Result 4
 218 also hold for the integer worst-phase counts we report.

219 *Result 4 (transfer across scenes).* Let m calibration scenes contribute $N =$
 220 $\sum_s n_s$ paired events, X of them discordant, and let J scenes show at least one
 221 discordant event. (a) *Population bound (the transfer guarantee).* If the scenes
 222 are drawn independently from a scene population (events within a scene may
 223 be dependent), the population-average discordance of a new camera is at
 224 most $U(J, m)$ with confidence $1 - \delta$, so its expected miss probability is at
 225 most its expected twin miss probability plus $U(J, m)$. The bound counts
 226 scenes, not events; it needs no independence of events within a scene. Its
 227 price is the number of scenes: with $J = 0$, $U \leq 10\%$ needs 29 scenes and

Table 4: Data needed for the per-camera form of Result 4(c) with no discordance ($J = 0$): minimum number of scenes $m_{\min}$, events per scene n at $m = m_{\min}$ and at $m = 2m_{\min}$, total events, and the asymptotic floor $\ln(1/\delta)/(\gamma t)$.

t	γ	$m_{\min}$	n at $m_{\min}$	n at $2m_{\min}$	total at $2m_{\min}$	floor
2.5%	5%	59	183	28, 3,304	2,397	
2.5%	10%	29	158	28, 1,624	1,198	
5%	5%	59	91	14, 1,652	1,198	
5%	10%	29	78	14, 812	599	
10%	5%	59	44	7, 826	599	
10%	10%	29	38	7, 406	300	

228 $U \leq 5\%$ needs 59. With the 76 fixed-camera scenes of this study, $J = 0, 1, 2, 3$
 229 give $U(J, m) = 3.9\%, 6.1\%, 8.1\%$ and 9.9% at level δ and $4.7\%, 7.1\%, 9.2\%$
 230 and 11.1% at level $\delta/2$: a single discordant scene costs about two percentage
 231 points, and a bound below $\varepsilon/2 = 10\%$ tolerates three discordant scenes at
 232 level δ but only two at $\delta/2$. Adding events to existing scenes does not tighten
 233 it; only new scenes do. (b) *Fleet bound (calibrated cameras, independent*
 234 *events)*. For the calibrated cameras, the event-weighted discordance is at
 235 most $U(X, N)$ with confidence $1 - \delta$ (for $\delta \leq 1/4$), however heterogeneous
 236 the cameras are; hence their event-weighted miss rate is at most their twin
 237 miss rate plus $U(X, N)$. Applying it requires a separate bound on the twin
 238 miss rate, with its own share of δ ; where real misses exist, direct Clopper–
 239 Pearson on them is at least as informative. The proof compares the Poisson–
 240 binomial count with a binomial of the same mean [50] and uses a lower
 241 bound on the probability that a binomial reaches its mean [51], applied to
 242 the complementary count. This bound assumes that events are independent
 243 given the scenes, so positive dependence within a scene makes it optimistic,
 244 and it says nothing about a new camera. (c) *Per-camera limit*. A per-camera
 245 statement, “ $p \leq q + t$ except on a set of cameras of mass γ ”, follows if every
 246 scene has at least $n_{\min}$ events, with $\gamma = U(J, m)/(1 - (1 - t)^{n_{\min}})$. Table 4
 247 turns this into a data requirement: for $\gamma = 5\%$ at least 59 scenes, each with
 248 tens to hundreds of events. No public benchmark we know offers this, so the
 249 per-camera form is reported as a limit.

250 *Demonstration procedure*. The procedure evaluated in Section 6 is: (1) de-
 251 clare the domain and pair every real event of m calibration scenes from it
 252 with its twin; count discordant scenes J ; (2) *domain rule*: if $U_{\delta}(J, m) > \varepsilon/2$,
 253 withdraw the transfer claim for the whole domain (a design rule for go/no-go,
 254 not a theorem); (3) for a new camera of the declared domain, run its twin

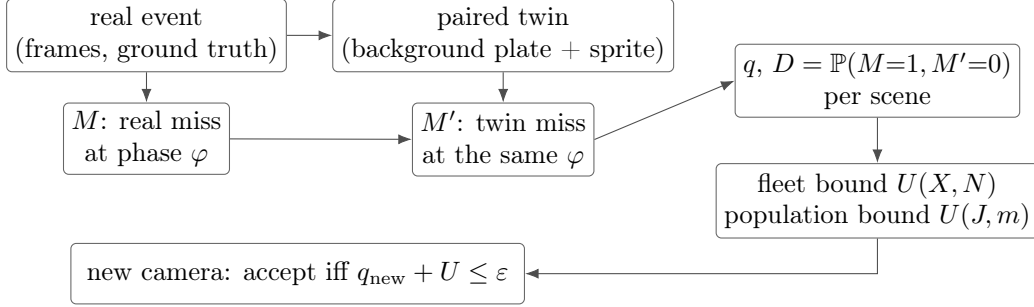


Figure 1: The demonstration with a paired twin. Each real event is paired with a synthetic twin that replaces the object and keeps the scene; both are evaluated by the same detector at the same sampling phase. Discordances calibrated across scenes bound the miss probability of a new camera through $p \leq q + D$.

255 until its own miss probability is bounded, q_U , at level $\delta/2$ (a twin *run* is one
 256 rendering of one synthetic event in one sprite variant; runs are treated as
 257 independent, which is optimistic when runs share a trajectory); (4) accept
 258 the camera with no real demand if $q_U + U_{\delta/2}(J, m) \leq \varepsilon$, otherwise fall back
 259 to the zero-failure test (Fig. 1). By Result 4(a) and a union bound, with
 260 confidence $1 - \delta$ the population-average discordance is at most $U_{\delta/2}(J, m)$.
 261 Among *accepted* cameras the average discordance can be larger, by at most
 262 the factor $1/\mathbb{P}(\text{accept})$, because acceptance selects cameras with a low twin
 263 miss rate; the average miss probability of accepted cameras is therefore at
 264 most $\varepsilon - U_{\delta/2} + U_{\delta/2}/\mathbb{P}(\text{accept})$. At OP*, where 97.4% of the static cam-
 265 eras are accepted, the inflation is 0.1 percentage points; out of domain it
 266 is unbounded (Section 6.3). The acceptance is *not* a per-camera guarantee.
 267 In the audit we nevertheless count, per camera, acceptances with $p_s > \varepsilon$
 268 (“false acceptances”) as a diagnostic of how the average guarantee behaves
 269 on individual cameras.

270 5. Experimental protocol

271 *Data.* The corpus is rebuilt from the raw detections and pixel ground truth
 272 of CDnet 2014 [52] and LASIESTA [53] (frames in true temporal order), with
 273 the synthetic BMC benchmark [49] as an off-the-shelf simulator; the detector-
 274 defined corpus has 3,170 events from 101 real videos. For the twin we use
 275 ground-truth events: a frame is a *hit* if a detector box covers at least half of
 276 the largest ground-truth component or overlaps its bounding box with IoU

277 ≥ 0.3 , and an event counts if its largest object has at least 256 pixels (primary
 278 definition; the original definition keeps every ground-truth event). Scenes are
 279 split into tiers before any twin result: a *static* (main) domain of 76 scenes (28
 280 CDnet, 48 LASIESTA; 211 events of any duration, 164 of at least 32 frames),
 281 a night tier (6 scenes), a turbulence tier (4 scenes; only 2 were run and they
 282 hold 2 events, so the tier is reported but not certified), a camera-jitter tier (2
 283 scenes of the CDnet category *cameraJitter*), and 4 pan-tilt-zoom scenes that
 284 are excluded because the twin assumes a static background. The camera-
 285 jitter tier is a correction made after the worst-phase and site-level checks of
 286 Section 6.5: in an earlier version of this study these scenes belonged to the
 287 static domain, and the checks placed its only fixed-phase false acceptance
 288 there. The split uses only the category label that CDnet already provides (a
 289 shaking camera breaks the static-background assumption, as pan-tilt-zoom
 290 does); no twin or detector was re-run, and every static-domain number below
 291 is recomputed without these scenes.

292 *Detector and twin.* The detector is a fixed single-stage segmentation network
 293 (YOLO family [54]; yolo26s-seg [55], confidence 0.25, input 640 px, CPU).
 294 Per scene, a background plate is the per-pixel median of up to 60 frames
 295 without ground-truth foreground. For each real event and each of three sprite
 296 variants (fixed seeds), the twin removes the real object (mask dilated by 3
 297 px), fills it from the plate, and draws a synthetic object on the ground-truth
 298 trajectory: a vehicle cut from Virtual KITTI 2 [56], chosen by aspect ratio,
 299 or a procedurally dressed human rendered with MPFB [57] with a four-frame
 300 walking cycle mirrored to the motion direction. The sprite height equals the
 301 ground-truth box height, its width is clamped to $\pm 10\%$ of the box width, its
 302 brightness follows the frame luminance, and a soft shadow is added. Only the
 303 object is synthetic; scene, illumination and background clutter are real. The
 304 detector runs on twin frames until the twin hits cover all residues modulo 48,
 305 which is exact for every K dividing 48. The median detector confidence on
 306 twin hits is 0.83 (real: 0.90). Building the twins of the 76 static scenes took
 307 16.5 CPU-hours (median 0.13 per scene) on a laptop CPU.

308 *What the twin consumes.* The paired twin uses, per camera, empty frames
 309 (for the plate) and the annotated trajectories of the camera’s real events (for
 310 sprite placement). Calibration therefore costs annotated events: 164 at OP*,
 311 i.e. 2.2 per static camera when amortized over the domain. For a *new* camera,
 312 the exchange rates below assume that the twin miss rate q_{new} can be measured

without that camera’s real events, e.g. with synthetic trajectories drawn on its empty background. We did not build such trajectory-free twins; we use the paired q_s in their place. The discordance of a trajectory-free twin has not been calibrated, so the reported savings are conditional on the paired-twin discordance carrying over, and are upper bounds.

Operating points. An operating point is $(\varepsilon, d_{\min}, K)$ with events of at least $d_{\min}$ frames. The primary point $\text{OP}^* = (20\%, 32, 16)$ at $\delta = 5\%$ was fixed by a rule chosen before the twin was built (the largest K for which direct certification needs at most 150 events at 80% power and 44 scenes give a population bound of at most $\varepsilon/2$), using the pooled real miss rate of all CDnet events (11.3% over 161 events, 112 events for direct certification). In the static domain the real miss rate at OP^* is only 0.61%; we report this, keep OP^* , and add the sensitivity points OP-A (20%, 32, 8) and OP-C (10%, 32, 1) and the full grid $d_{\min} \in \{1, 4, 8, 12, 16, 24, 32\}$, $K \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$. The night and turbulence tuning criteria and their held-out scenes were fixed before tuning. These choices are dated in internal project records but were not publicly registered. Five choices were made after results were available and are disclosed: the primary event definition (largest object at least 256 pixels) was adopted after the initial twin run, with the original definition kept as a co-reported sensitivity (Table 12); the LASIESTA scene list was rebuilt (20 to 48 scenes) after correcting the frame order of that dataset, before the static-domain twin was built; the two held-out turbulence scenes were not run; the post-hoc point (16, 16) of Section 6 was selected after inspecting the static-domain results; and the camera-jitter scenes were moved out of the static domain after the fixed-phase and site-level checks (above). The operating point OP^* , the other tiers, the audit design and the night criterion were fixed before the static-domain twin results; everything else in Section 6 is exploratory.

Why a paired twin. A reweighting alternative [58] needs overlap between synthetic and real events. A classifier separating BMC events from real events reaches a median AUC of 0.64 with temporal features, 0.94 with geometry and 0.77 with detector confidence alone (all targets); the gap lies mostly in geometry, so a reweighted synthetic benchmark cannot stand in for the real event distribution, and the twin is matched on geometry by construction.

Audit. Leave-one-scene-out (LOSO): each scene is certified with the bound computed from the other scenes. Monte Carlo (10,000 draws): calibration

Table 5: Certificates by tier at OP* ($\delta = 5\%$). Discordant events are counted at the worst sampling phase (in parentheses: phase-averaged count, mean over the three sprite variants; per-scene tables use the majority rule, which gives 8.31 at night); $\hat{p}$, $\hat{q}$: real and twin miss rates averaged over the phases / at the worst phase (any fixed deployment phase); J = scenes with a discordant event.

tier	events	$\hat{p}$ avg/worst	$\hat{q}$ avg/worst	discordant	$U(X, N)$	J/m	$U(J, m)$
static (main)	164	0.6%/0.6%	1.0%/2.4%	0 (0.0000)	1.81%	0/76	3.87%
camera jitter	16	2.0%/18.8%	0.0%/0.0%	3	41.7%	1/2	97.5%
night	25	36.2%/52.0%	4.8%/16.0%	12 (8.21)	65.9%	5/6	99.1%
turbulence	2	6.2%/50.0%	0.0%/0.0%	1	97.5%	1/2	97.5%
all tiers	207	5.1%/8.7%	1.3%/3.9%	16	11.5%	7/86	14.7%

349 scenes are resampled from the static domain and a new camera is drawn from
350 it; a false certificate is a certified camera with $p_s > \varepsilon$ (the resampled calibration
351 set may contain the test camera, which makes this check optimistic; the
352 LOSO audit does not have this defect); the “truth” of a camera is its own
353 event pool, exact over phases. Two must-fail scenarios are run (Section 6.3).
354 All randomness uses seed 42.

355 6. Results

356 6.1. Certificates by tier

357 In the static domain, 0 of 164 events at OP* are discordant; hence
358 $U(X, N) = 1.81\%$ and, with $J = 0$ of 76 scenes, $U(J, m) = 3.87\%$ (Ta-
359 ble 5, Fig. 2). Both are below $\varepsilon/2 = 10\%$. In the camera-jitter tier, 3 of 16
360 events are discordant, all in one scene. At night, 12 of 25 events are discor-
361 dant (48%; phase-averaged 8.21) and 5 of 6 scenes; pooling all tiers raises
362 the bounds to 11.5% and 14.7%.

363 6.2. Exchange rate

364 Fig. 3 and Table 7 show the real events a static camera saves when the
365 twin replaces direct certification, across 70 operating points. A camera is
366 certified with no real event when its twin miss rate plus the transfer bound
367 (both at $\delta/2$, the bound computed without the camera) stays below ε . At
368 OP* the direct test needs only $n_0 = 14$ events, because the static-domain
369 miss rate is low; a twin run to convergence saves those 14 events at 74 of 76
370 cameras, at a measured cost of 0.037 CPU-hours per saved event (conditional
371 on the discordance carrying over, Section 5; no accepted camera has $p_s > \varepsilon$

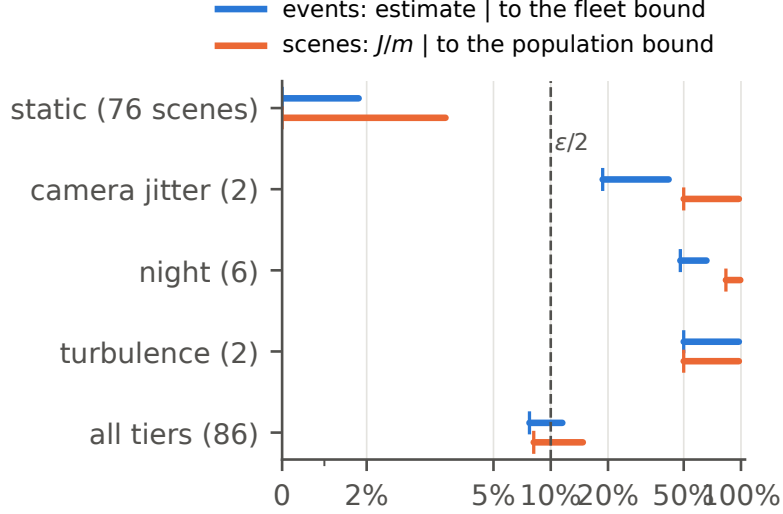


Figure 2: Certificates by tier at OP^* (event unit, worst phase). Blue: discordant events per event and the fleet bound; orange: discordant scenes per scene and the population bound. Dashed: $\varepsilon/2$.

372 at OP^* , but acceptances were not audited at every grid cell). **The twin as**
 373 **built saves no real events; the saving requires about 23 twin runs**
 374 **per camera.** With three runs per event, as built, the twin's own miss rate
 375 cannot be bounded tightly enough, and at most 5 cameras save anything
 376 at any grid point. The saving grows with the real miss rate, because direct
 377 certification becomes expensive: it reaches hundreds of events per camera
 378 when $\hat{p}$ exceeds 10%. At the post-hoc point (16, 16) it is 22 events per camera
 379 (Table 6). The target $\varepsilon = 20\%$ is a demonstrator, not a safety target. At
 380 $\varepsilon = 5\%$ the transfer term alone is $U_{\delta/2}(0, m) = 4.7\%$ even with no discordant
 381 scene, leaving 0.3% for the twin's own miss rate, i.e. at least 1406 miss-free
 382 twin runs per camera; with 76 scenes the twin route is therefore impractical
 383 at $\varepsilon = 5\%$, and 59 or more scenes with no discordance are needed before
 384 it can work. For the calibrated cameras themselves, direct Clopper–Pearson
 385 on their pooled real misses gives 2.9%, close to the fleet discordance bound
 386 1.81%: the fleet certificate adds little where real data exist, and its role is as
 387 the building block of the transfer.

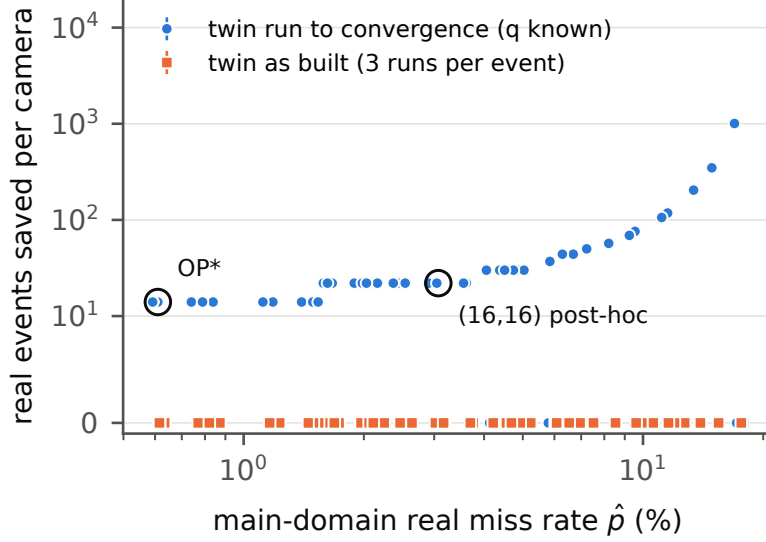


Figure 3: Exchange rate: real events saved per static camera (median and interquartile range over 76 cameras) against the static-domain real miss rate, one point per $(d_{\min}, K)$ at $\varepsilon = 20\%$. Blue: twin run to convergence; orange: twin as built (three runs per event). Vertical axis: symmetric log (linear below 10). Circled: OP^* and the post-hoc point (16, 16).

6.3. Audit and must-fail scenarios

Is the certificate right because it is empty? Table 8 cross-tabulates real and twin outcomes. In the static domain at OP^* almost every event is caught by both: 1 events have a real miss at some phase, in 1 scenes, and the twin also misses 100% of the expected real misses, i.e. one event. The zero-discordance result is therefore mostly a statement about a detector that rarely misses in this domain, not a test of the twin. At the post-hoc point the twin sees most real misses; at night it sees almost none.

LOSO and Monte Carlo. At OP^* , the number of static scenes whose discordance exceeds the bound computed without them is 0 of 76 (Fig. 5, Table 9); the Monte-Carlo false-certification rate is 0.0%. Because real misses are rare at OP^* , this audit cannot reveal much: no static scene has $p_s > \varepsilon$ there, so a false certificate is impossible by construction and the 0% rate carries no evidence. A scene above its leave-one-out bound would still be compatible with a population-average guarantee. The informative tests are

Table 6: Exchange rate at four operating points (76 static cameras). n_{dir} : real events of the smallest direct plan (failures allowed, Clopper–Pearson acceptance) with 80% power at the pooled static-domain $\hat{p}$; the zero-failure plan alone needs n_0 events; saved: median real events saved per camera by a twin run to convergence; runs: twin runs needed per camera; CPU-h: CPU-hours per saved real event.

point	$\hat{p}$	n_{dir}	saved	cams	runs	CPU-h
OP* (20%, 32, 16)	0.61%	14	14	74	23	0.037
OP-A (20%, 32, 8)	0.61%	14	14	75	23	0.035
OP-C (10%, 32, 1)	0.61%	29	29	75	70	0.049
(16, 16) post hoc	3.07%	22	22	73	33	0.034
worst-phase convention (valid for any fixed phase):						
OP*	0.6%	14	14	72	23	–
(16, 16) post hoc	7.3%	50	50	71	33	–
twin as built: cameras saving = 3, 3, 0, 2 (median saving 0)						

403 must-fail 2 and the camera-jitter check below, which can fail and do.

404 *Must-fail 1: temporal.* At $K = 48$, 3 static scenes have $p_s > \varepsilon$ and the
405 procedure refuses them in 100% of runs; but most of these misses are temporal
406 (events shorter than K , which the twin reproduces by construction); the
407 number of scenes that exceed ε through detector misses is 1 (Table 10, top).

408 *Must-fail 2: detector.* We add k of the 6 night scenes to the calibration
409 as if they were static-domain scenes; at OP* and (16, 16) every real miss
410 of these scenes is a detector miss. The transfer bound rises with k (Fig. 4,
411 Table 10, bottom): from 3.9% to 12.4% at OP*, crossing $\varepsilon/2$ for some subsets
412 at $k = 4$ and for all at $k = 5$, so the domain-level certificate is withdrawn; no
413 static camera is ever falsely certified (0). *Out-of-domain pass-through:* the
414 per-camera rule does not protect the night cameras themselves. The night
415 twin misses almost nothing (its appearance model only dims the sprite), so
416 each night camera, judged with a bound computed from the static scenes,
417 is certified; 4 of 6 night cameras have $p_s > \varepsilon$ and are falsely certified at
418 OP* (4 at (16, 16); Table 10, middle). Each night miss rate rests on 2 to
419 10 events and is uncertain; the finding is that the acceptance rule admitted
420 these cameras although their observed miss rates exceed ε and their twins
421 miss almost nothing, not a precise estimate of their miss rates.

422 *Camera jitter.* The same pass-through occurs for a shaking camera.
423 Judged by the twin route calibrated on the static scenes (worst-phase con-

Table 7: Operating-point grid at $\varepsilon = 20\%$ (condensed; full grid in the supplement). Each cell: static-domain real miss rate $\hat{p}$ / real events for direct certification / median real events saved per camera by a twin run to convergence. The twin as built saves none at every cell. OP* is (32, 16).

$d_{\min} \backslash K$	1	8	16	24	48
32	0.6%/14/14	0.6%/14/14	0.6%/14/14	0.8%/14/14	4.1%/30/0
24	0.6%/14/14	0.7%/14/14	1.2%/14/14	1.6%/22/22	5.8%/37/0
16	1.1%/14/14	1.9%/22/22	3.1%/22/22	4.4%/30/30	9.7%/76/0
12	1.7%/22/22	2.4%/22/22	3.6%/22/22	4.9%/30/30	10.2%/88/0
8	1.6%/22/22	2.9%/22/22	5.0%/30/30	6.7%/44/44	12.2%/141/0
4	2.0%/22/22	6.3%/44/44	9.5%/76/76	11.5%/118/118	17.2%/1164/0
1	2.4%/22/22	11.1%/106/106	14.9%/348/348	16.9%/1006/1006	22.4%/–/0

Table 8: Real versus twin outcomes (expected event counts under a uniform sampling phase; twin = majority of three variants). Recall: share of expected real misses that the twin also misses. ^{ph} post hoc.

	static, OP*	static, (16, 16) ^{ph}	night, OP*
real miss, twin miss	1.00	4.81	0.75
real miss, twin catch (D)	0.00	0.69	8.31
real catch, twin miss	0.56	3.62	0.44
real catch, twin catch	162.44	169.88	15.50
events	164	179	25
events missed at some phase	1	13	13
twin recall of real misses	100%	88%	8%

424 vention, Section 6.5), 2 of the 2 camera-jitter cameras are accepted and 1
425 of them, whose worst-phase real miss rate is 37.5%, is falsely accepted (1 at
426 (16, 16), worst-phase real miss rate 53.8%; per-camera table in the supple-
427 ment). Its twin catches every event at every phase, because the synthetic
428 object is drawn on a stabilised plate while the real frames shake; under the
429 phase-averaged reading this camera stays below ε (Table 10, camera-jitter
430 rows), so this failure appears only for a fixed deployment phase. Under the
431 same worst-phase rule 4 of 6 night cameras are falsely accepted at OP*.

432 6.4. Illustrative example at a post-hoc operating point

433 To see the certificate under actual real misses, we moved, after inspecting
434 the data, to $(d_{\min}, K) = (16, 16)$, the nearest operating point with at least 20
435 missed real events in the earlier, wider static domain. This choice is post hoc
436 and is reported as an illustration only. Without the camera-jitter scenes, 13
437 of 179 events have a real miss at some phase and the twin also misses 13 of
438 them (counting events); weighting by phases, the twin reproduces 88% of the

Table 9: Leave-one-scene-out: scenes whose discordance D_s exceeds the bound U_{-s} computed without them. None is falsely certified, because $p_s \leq \varepsilon$.

point	scene	n	D_s	p_s	q_s	U_{-s}	$p_s > q_s + U_{-s}$
OP*	none						
(16, 16) ^{ph}	tramCrossroad_1fps	4	9.4%	12.5%	6.2%	6.2%	yes

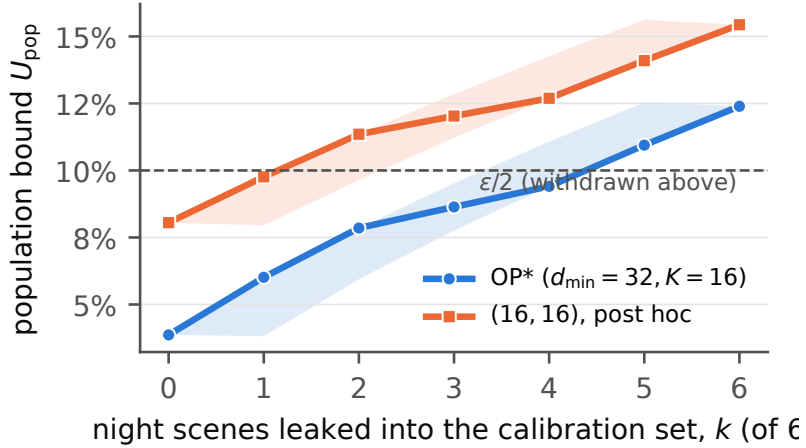


Figure 4: Must-fail 2: the population bound when k night scenes leak into the static calibration set (median over the $\binom{6}{k}$ subsets, band: min-max). The domain certificate is withdrawn above $\varepsilon/2$. Individual night cameras are still certified when judged alone (out-of-domain pass-through, text).

439 expected real misses; $J = 2$, $U(J, m) = 8.1\%$, the number of scenes above
 440 their LOSO bound is 1 of 76, and no camera is falsely certified (0). Before
 441 looking at the recomputed numbers we required, for keeping this point after
 442 the domain correction, both bounds below $\varepsilon/2$ and no false acceptance under
 443 the worst-phase convention; it meets all three: 5 of 179 events discordant at
 444 the worst phase, $U(X, N) = 5.8\%$, $U(J, m) = 8.1\%$, and 0 false acceptances
 445 among 71 accepted cameras (Table 6).

446 6.5. Fixed-phase and site-level checks

447 *Worst phase.* A deployed camera keeps one sampling phase. Counting
 448 an event as missed if it is missed at any phase, for the real event, the twin
 449 (majority of variants) and the discordance alike, keeps the bound valid for
 450 every fixed phase, because an event missed at some phase is either missed

Table 10: Must-fail scenarios. Top: temporal ($d_{\min} = 1$, $K = 48$), static scenes with $p_s > \varepsilon$, real misses split into detector and temporal misses; all refused in every Monte-Carlo run. Middle: night scenes at OP* (every miss is a detector miss), and whether each is falsely certified when judged alone (phase-averaged); then the same for the camera-jitter scenes. Bottom: population bound when k night scenes leak into the calibration set (median over subsets; withdrawn = share of subsets with a bound above $\varepsilon/2$).

<i>1: temporal</i>						
scene	n	p_s	detector	temporal	q_s	refused
port_0_17fps	5	99.6%	43.3%	56.2%	100.0%	100%
tramCrossroad_1fps	11	77.7%	6.1%	71.6%	74.2%	100%
tunnelExit_0_35fps	34	83.6%	5.2%	78.4%	87.9%	100%
<i>2: detector (night)</i>						
scene	n	p_s	q_s	D_s	disc.	false cert.
bridgeEntry	2	15.6%	0.0%	15.6%	1	no
busyBoulevard	4	96.9%	0.0%	96.9%	4	yes
fluidHighway	4	26.6%	0.0%	26.6%	2	yes
streetCornerAtNight	10	28.1%	11.9%	20.6%	4	yes
tramStation	2	0.0%	0.0%	0.0%	0	no
winterStreet	3	33.3%	0.0%	33.3%	1	yes
<i>3: detector (camera jitter)</i>						
boulevard	8	0.0%	0.0%	0.0%	0	no
traffic	8	3.9%	0.0%	3.9%	3	no
k leaked	0	1	2	3	4	6
U_{pop}	3.9%	6.0%	7.9%	8.6%	9.4%	12.4%
withdrawn	0%	0%	0%	0%	33%	100%

by the twin or discordant at that phase. At OP* the static-domain real and twin miss rates become 0.6% and 2.4% (phase-averaged 0.6% and 1.0%), while the discordance counts and both bounds are unchanged (Table 5). The direct plan then needs 14 events, and a twin run to convergence accepts 72 of 76 cameras, saving 14 events each (Table 6); the number of accepted static cameras with a worst-phase real miss rate above ε is 0 (0 at (16, 16)). In the earlier version of this study the camera-jitter scene *traffic* was in the static domain and was such a false acceptance; it is now reported as a failure mode (Section 6.3).

Sites. Scenes filmed at the same site share background, viewpoint and illumination, so the scene may be too fine a unit of independence. Grouping the static scenes into 21 sites (CDnet challenge categories and LASIESTA sequence groups, a coarse proxy for sites), the number of sites with a discordant event at OP* is 0, yet with so few units the population bound is $U(J_{\text{site}}, m_{\text{site}}) = 13.3\%$, above $\varepsilon/2$: under the domain rule the transfer claim

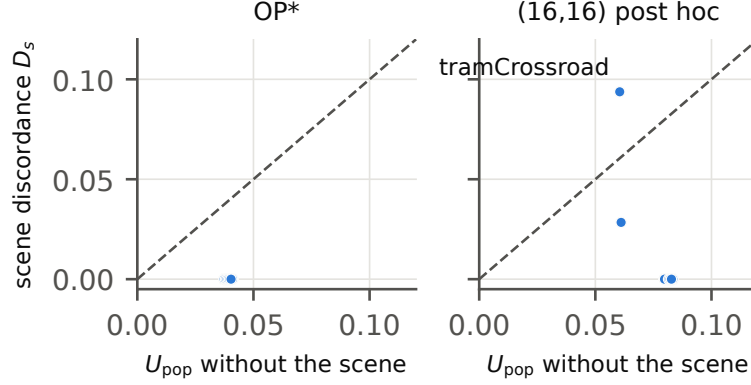


Figure 5: Leave-one-scene-out: per-scene discordance against the population bound computed without the scene. Points above the diagonal exceed the bound; the proposition bounds the population mean, not each scene.

466 is withdrawn at site level. At (16,16) it is 20.7% (1 of 21 sites). The leave-
 467 one-site-out audit finds 0 sites above their bound at OP* and 1 at (16,16).
 468 The transfer certificate therefore stands at scene level and is withdrawn at
 469 site level; it holds only if scenes, not sites, are the independent units.

470 6.6. Failure modes

471 At night the twin sees few real misses: it predicts 8% of the expected
 472 missed events, and the tier bound is 99.1%. Fig. 6 shows why: the appearance
 473 model scales sprite brightness with the scene, but has no headlights, glare or
 474 sensor noise. An attempt to fix this was specified in advance: night-specific
 475 appearance profiles were tuned on four night scenes and judged on two held-
 476 out scenes, with success defined as a median held-out per-scene discordance
 477 of at most 5%. The best profile barely moved the tuning discordance (47.2%
 478 to 47.0%, phase-averaged) and the held-out median was 23.2% on 14 events,
 479 so the criterion failed, tuning was stopped, and night is declared outside the
 480 domain. Camera jitter is the second named failure mode: the twin is drawn
 481 on a stabilised background plate, so it cannot reproduce the misses caused by
 482 camera shake (twin worst-phase miss rate 0.0% against a real 18.8% at OP*;
 483 Table 5). The turbulence tier holds 2 events at OP* and is not certified.



Figure 6: Synthetic person sprites (CC0) on a neutral background under the brightness gain used for day and for night scenes. Dimming alone does not reproduce night imaging, which explains the low twin miss rate at night.

484 6.7. Sensitivity

485 *Event definition.* Keeping every ground-truth event (273 instead of 211
486 static events of any duration) leaves OP^* almost unchanged: 166 events, 0
487 discordant, $U(X, N) = 1.79\%$, $J = 0$ (Table 12). At the post-hoc point the
488 original definition adds small, hard objects, and the population bound rises
489 to 9.9% with $J = 3$, at the edge of $\varepsilon/2$: the illustrative example is sensitive
490 to the definition, while OP^* is not. *Operating point.* At OP-A and OP-C the
491 static-domain miss rate is 0.61% and 0.61%, and no static-domain event is
492 discordant. *Twin details.* The majority rule over sprite variants (1, 2 or 3 of
493 3) does not change any static-domain count. Table 11 compares twin and real
494 hit rates on the same frames: the twin is slightly pessimistic at every aspect
495 error except one vehicle bin, and both hit rates fall when the ground-truth
496 box has an unusual aspect (occlusion, objects entering the frame). Clamping
497 the sprite width, which affects 70% of twin frames (more than the frames
498 with an aspect error above 10% in Table 11, because 1-pixel rounding also
499 triggers the clamp), lowers the twin hit rate by only 3.0 points more than
500 the real hit rate on the same frames (631 event-variant pairs): it inflates q ,
501 not D . The median detector confidence on hits is lower for the twin than for
502 real objects, especially for persons. Full tables are in the supplement.

Table 11: Twin fidelity on the same frames (static domain): hit rates by object kind and sprite aspect error before the width clamp, and median detector confidence on hits. Negative differences make the twin pessimistic.

kind	aspect error	frames	twin hit	real hit	diff. (pts)
person	$\leq 10\%$	8,870	91.7%	94.8%	-3.2
person	10-20%	5,899	90.1%	94.2%	-4.2
person	20-40%	4,404	74.0%	86.2%	-12.2
person	$> 40\%$	5,449	34.4%	44.5%	-10.0
vehicle	$\leq 10\%$	4,105	86.9%	89.0%	-2.1
vehicle	10-20%	2,569	88.0%	90.2%	-2.2
vehicle	20-40%	3,087	87.6%	85.1%	+2.5
vehicle	$> 40\%$	3,881	64.9%	74.2%	-9.3
person	confidence		0.86	0.92	
vehicle	confidence		0.72	0.78	

Table 12: Event-definition sensitivity (static domain). Primary: largest object ≥ 256 px; original: every ground-truth event. Discordant events at the worst phase; LOSO: scenes above the bound computed without them.

definition	point	N	$\hat{p}$	disc.	$U(X, N)$	$J / U(J, m)$	LOSO
primary	OP*	164	0.61%	0	1.81%	0 / 3.9%	0
	OP-A	164	0.61%	0	1.81%	0 / 3.9%	0
	OP-C	164	0.61%	0	1.81%	0 / 3.9%	0
	(16, 16) ^{ph}	179	3.07%	5	5.78%	2 / 8.1%	1
original	OP*	166	0.60%	0	1.79%	0 / 3.9%	0
	OP-A	166	0.60%	0	1.79%	0 / 3.9%	0
	OP-C	166	0.60%	0	1.79%	0 / 3.9%	0
	(16, 16) ^{ph}	184	4.59%	6	6.33%	3 / 9.9%	1

^{ph} post-hoc point.

7. Discussion and limitations

A narrow window. Simulation pays in events only when direct certification is expensive, that is, when the real miss rate is not far below ε ; there the twin must also be accurate, and discordances make the transfer bound grow. When the miss rate is low, n_0 is small and there is little to save. The grid (Table 7) shows both ends: 14 events at OP*, hundreds at operating points where $\hat{p}$ approaches ε .

Time. The waiting time saved at camera s is $n_{\text{saved}}/\lambda_s$, where λ_s is the event rate in deployment; we do not estimate it, because the benchmarks publish no frame rates and their clips are dense in events. If one CPU-hour is valued like one hour of waiting, the twin breaks even when $\lambda_s < 1/c \approx 27$ events per hour at OP*, with c the CPU-hours per saved event; this

515 equivalence is an assumption, not a measurement.

516 *Domain.* The transfer bound is a population-average guarantee: it with-
517 draws the domain certificate when out-of-domain scenes enter the calibration,
518 but it does not detect an out-of-domain camera. The domain must therefore
519 be declared and checked by other means (for example, illumination statis-
520 tics), and a per-camera guarantee requires the data of Result 4(c). Three
521 items are left for future work: a twin that needs no annotated real trajec-
522 tory of the new camera, with its own calibrated discordance; a pre-specified
523 in-domain stress test with cameras whose real miss rate exceeds ε ; and a
524 domain check (for example a per-camera discordance sentinel on a few real
525 events) to stop out-of-domain pass-through.

526 *Data and twin.* No static scene has 59 or more qualifying events, so per-
527 camera ground truth is the scene’s own finite event pool; the twin builds its
528 plate from real empty frames and therefore needs a static background; the
529 sprite library lacks elevated viewpoints for vehicles; and we did not use a full
530 3-D simulator. The operating point was fixed from the pooled miss rate of all
531 CDnet events, which turned out to be much higher than the static-domain
532 rate; we kept it rather than re-select after seeing the twin results. Scenes are
533 not independent draws: several LASIESTA sequences share a room or set-
534 up, and the site-level bound of Section 6.5 exceeds $\varepsilon/2$. The cost comparison
535 counts real events and CPU-hours only; annotating the calibration events and
536 building plates are additional costs. Finally, an event miss is a component-
537 level failure; linking it to a system hazard requires a hazard analysis of the
538 application.

539 *Independence and phase.* The results assume independent events given
540 the scene and independent scenes. Events of one camera share a background,
541 a detector and, in deployment, a single sampling phase; benchmark scenes are
542 not a random sample of deployments. Correlation of events within a scene
543 does not invalidate the population bound of Result 4(a) or the leave-one-
544 scene-out audit, because both count scenes, not events: a scene contributes
545 one indicator whatever the dependence among its events. It does affect the
546 fleet bound of Result 4(b), which counts events and becomes optimistic under
547 positive within-scene dependence; the population bound needs only indepen-
548 dence across scenes. The main results certify the phase-averaged miss rate;
549 the worst-phase convention of Section 6.5 covers any fixed phase at a higher
550 real-event cost.

551 *Threats to validity.* The audit tables and figures use level δ , while ac-
552 ceptance uses $\delta/2$ for the transfer term; at $\delta/2$ the bounds are larger (Re-

553 sult 4(a)). Software: Python 3.11.9, NumPy 1.26.4, SciPy 1.16.3, pandas
554 2.3.3, Ultralytics 8.4.7, OpenCV 4.10.0; CPU: Intel Core i7-1185G7. The
555 detector, the event definition and the hit rule are fixed; a different detector
556 changes p , q and D but not the theorems. The number of scenes, not the
557 number of events, limits the transfer certificate: the static domain reaches
558 $U(J, m) = 3.87\%$ with no discordant scene among 76, and more scenes are
559 needed for tighter bounds. The audit at OP* is weak because real misses are
560 rare there; the post-hoc point shows the mechanism under real misses but was
561 selected after inspecting the data, and its population bound is sensitive to
562 the event definition (Table 12). The static domain itself was narrowed after
563 the fixed-phase checks (camera jitter, Section 5); its clean audit is therefore
564 partly a result of that post-hoc correction.

565 *Practical guidance.* The results suggest a simple decision rule for practi-
566 tioners who consider replacing a field demonstration with a twin. (i) Compute
567 n_0 at the target (ε, δ) and estimate the real miss rate; if it is far below ε , the
568 direct test is already short and a twin can save at most n_0 events. (ii) If a
569 twin is built, declare its domain before calibration and calibrate discordance
570 on as many scenes of that domain as possible; the number of scenes sets the
571 resolution (Result 4(a)). (iii) Run each new camera’s twin long enough to
572 bound its own miss rate; three runs per event, as here, are not enough. (iv)
573 Do not use the transfer certificate to admit a camera whose conditions dif-
574 fer from the calibration domain (night, a shaking mount): the certificate is
575 withdrawn when such scenes enter the calibration, but a single such camera
576 passes through.

577 8. Conclusion

578 Without an assumption linking simulation to reality, synthetic evidence
579 buys no real events in zero-failure certification, and relaxing the error level
580 buys an exactly computable number. A paired twin turns the question into
581 one of discordance. The population bound is the transfer guarantee: it counts
582 scenes, tolerates dependence within a scene, and its price is the number of
583 scenes; the fleet bound covers only the calibrated cameras and assumes inde-
584 pendent events. On the 76 static scenes of this study the transfer certificate
585 stands at scene level ($U(J, m) = 3.87\%$) but is withdrawn at site level, where
586 21 sites give 13.3%, above $\varepsilon/2$. Measured on public benchmarks, the ex-
587 change rate is modest and domain-bound, and the certificate has two named

588 failure modes, night and camera jitter, in which out-of-domain cameras pass
589 through unless the domain is declared.

590 **CRediT authorship contribution statement**

591 **Lam Quoc Dat:** Conceptualization, Methodology, Software, Validation,
592 Formal analysis, Investigation, Data curation, Writing – original draft, Writ-
593 ing – review & editing, Visualization.

594 **Declaration of competing interest**

595 The author declares no competing interests. This research received no
596 specific grant from any funding agency in the public, commercial, or not-for-
597 profit sectors.

598 **Declaration of generative AI and AI-assisted technologies in the** 599 **manuscript preparation process**

600 During the preparation of this work the author used an AI assistant
601 (Claude, Anthropic) to write analysis and plotting scripts, to check the num-
602 bers of the manuscript against the result files, and to draft and edit text.
603 After using this tool, the author reviewed and edited the content as needed
604 and takes full responsibility for the content of the publication.

605 **Data availability**

606 Code, event manifests, result tables and the CC0 person-sprite set are
607 publicly available at [https://github.com/lamquocdat01/no-free-credi](https://github.com/lamquocdat01/no-free-credit-ress-artifact)
608 [t-ress-artifact](https://github.com/lamquocdat01/no-free-credit-ress-artifact) (tag `ress-v1.1`, commit `22da0795de09`). Raw video frames
609 (CDnet 2014, LASIESTA, BMC) and Virtual KITTI 2 sprites are not redis-
610 tributed under their licences; the repository provides download instructions
611 and SHA-256 checksums to regenerate every derived file.

612 **Appendix A. Formal statements and proofs**

613 This appendix states the results of Section 4 formally and proves them.
614 Theorem A.1 is Result 1, Theorem A.3 is Result 2, Lemma A.5 and Propo-
615 sition A.15 are Result 3, and Proposition A.11, Theorem A.9 and Proposi-
616 tion A.13 are Result 4(a), (b) and (c).

617 *Appendix A.1. Setting and notation*

618 **Events and misses.** A camera (scene) s produces events $i = 1, 2, \dots$ (a
619 maximal run of frames containing the target object). The system samples
620 frames periodically with period K and an unknown phase $\varphi \in \{0, \dots, K-1\}$;
621 a detector is run on the sampled frames. An event is *missed* ($M = 1$)
622 when no sampled frame of the event is a hit (the detector finds the object).
623 With $H \subseteq \mathbb{Z}$ the hit frames of the event and φ uniform, $\mathbb{P}(M = 1 \mid H) =$
624 $1 - |H \bmod K|/K$, where $H \bmod K$ is the set of residues of H . The miss
625 rate of camera s is $p_s = \mathbb{P}(M = 1)$ over its events and phases. Target: certify
626 $p_s \leq \varepsilon$ with confidence $1 - \delta$.

627 **Clopper–Pearson bound.** For $x \in \{0, \dots, n\}$, $F_{n,p}(x) = \mathbb{P}(\text{Bin}(n, p) \leq x)$
628 and $U(x, n) = U_\delta(x, n)$ is the one-sided $(1 - \delta)$ upper Clopper–Pearson bound
629 [59]: the solution u of $F_{n,u}(x) = \delta$ for $x < n$, and 1 for $x = n$. Since $F_{n,p}(x)$
630 is strictly decreasing in p , $U(x, n) < p \iff F_{n,p}(x) < \delta$, and U is non-
631 decreasing in x . Validity: $\mathbb{P}(U(\text{Bin}(n, p), n) < p) \leq \delta$ for every p .

632 **Success run.** $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1 - \varepsilon) \rceil$ is the smallest n with $(1 - \varepsilon)^n \leq \delta$
633 (e.g. $n_0(5\%, 5\%) = 59$, $n_0(20\%, 5\%) = 14$).

634 **Simulator and twin.** Simulator data Y have a law Q on an arbitrary space.
635 A *twin* is a paired simulator: every real event i has a synthetic counterpart
636 with miss indicator M'_i under the same sampling phase. $q = \mathbb{P}(M' = 1)$, and
637 the *discordance* is $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch).

638 *Appendix A.2. C1: no free credit*

639 A certification procedure is a (possibly randomised) test $\phi(X, Y) \in \{0, 1\}$
640 with real data $X = (M_1, \dots, M_n) \sim \text{Bern}(p)^{\otimes n}$ independent of simula-
641 tor data $Y \sim Q$. It is *valid at level δ on a class $\mathcal{P}$* of pairs (p, Q) if
642 $\sup_{(p, Q) \in \mathcal{P}, p > \varepsilon} \mathbb{P}_{p, Q}(\phi = 1) \leq \delta$.

643 **Theorem A.1 (C1).** Assume $\mathcal{P}$ places no restriction linking Q and p : for
644 every Q there are $(0, Q) \in \mathcal{P}$ and $(p, Q) \in \mathcal{P}$ for all $p \in (\varepsilon, \varepsilon + \eta)$ for some
645 $\eta > 0$. If ϕ is valid at level δ on $\mathcal{P}$, then for every Q and every simulator
646 sample size,

$$(1 - \varepsilon)^n \mathbb{P}_{0, Q}(\phi = 1) \leq \delta.$$

647 In particular, certifying a perfect system ($p = 0$) with probability one requires
648 $n \geq n_0(\varepsilon, \delta)$ real events, whatever the amount of simulator data; power $1 - \beta$
649 at $p = 0$ requires $n \geq \ln(\delta/(1 - \beta))/\ln(1 - \varepsilon)$.

650 *Proof.* Fix Q and write $g(y) = \mathbb{E}[\phi(0^n, y)]$ (expectation over the internal
651 randomisation). Under $p = 0$, $X = 0^n$ almost surely, so $\mathbb{P}_{0,Q}(\phi = 1) =$
652 $\mathbb{E}_Q[g(Y)]$. Under $p > \varepsilon$, the event $\{X = 0^n\}$ has probability $(1 - p)^n$ and is
653 independent of Y , hence $\mathbb{P}_{p,Q}(\phi = 1) \geq (1 - p)^n \mathbb{E}_Q[g(Y)] = (1 - p)^n \mathbb{P}_{0,Q}(\phi =$
654 $1)$. Validity gives $(1 - p)^n \mathbb{P}_{0,Q}(\phi = 1) \leq \delta$ for all $p \in (\varepsilon, \varepsilon + \eta)$; letting $p \downarrow \varepsilon$
655 proves the inequality. With $\mathbb{P}_{0,Q}(\phi = 1) = 1$ it reads $(1 - \varepsilon)^n \leq \delta$, i.e. $n \geq n_0$;
656 with $\mathbb{P}_{0,Q}(\phi = 1) \geq 1 - \beta$ it gives the power bound. $\square$

657 *Remark A.2.* The argument only uses that the simulator data have the same
658 law under the two hypotheses and that the all-success real sample has positive
659 probability under both. It is the success-run analogue of the observation
660 that external data cannot increase power under strict type-I error control
661 over all external-data laws [19]. *Check.* The zero-failure test with $n = 59$ at
662 $p = \varepsilon = 5\%$: $(0.95)^{59} = 0.0485$ (MC 0.0491, 10^5 runs).

663 *Appendix A.3. C2: the price of borrowing*

664 Suppose one accepts a two-tier guarantee: level δ when the simulator is
665 “right” and only level $\delta' > \delta$ in general.

666 **Theorem A.3** (C2, exact credit). *(i) Converse. Under the assumption of*
667 *Theorem A.1, any procedure valid at level δ' on $\mathcal{P}$ that certifies a perfect*
668 *system with probability one uses $n \geq n_0(\varepsilon, \delta')$ real events. Hence the credit,*
669 *the number of real events saved with respect to $n_0(\varepsilon, \delta)$, is at most*

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta').$$

670 *(ii) Attainment. Let $T(Y) \in \{0, 1\}$ be any simulator test (“the simulator*
671 *passes”) and let B be the procedure: if $T = 1$ run the zero-failure test with*
672 *$n_1 = n_0(\varepsilon, \delta')$ real events, otherwise with $n_0 = n_0(\varepsilon, \delta)$. Then B is valid at*
673 *level δ' on $\mathcal{P}$, saves exactly c events whenever $T = 1$, and is valid at level δ*
674 *on the subclass*

$$\mathcal{P}_0 = \left\{ (p, Q) : p > \varepsilon \Rightarrow \mathbb{P}_Q(T = 1) \leq \pi^* \right\}, \quad \pi^* = \frac{\delta - (1 - \varepsilon)^{n_0}}{(1 - \varepsilon)^{n_1} - (1 - \varepsilon)^{n_0}},$$

675 *i.e. where a simulator of a system that is truly worse than ε rarely passes.*

676 *(iii) Continuous lower bound. $c \geq c^* = \lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$.*

677 *Proof.* (i) Theorem A.1 at level δ' gives $(1 - \varepsilon)^n \leq \delta'$, whose smallest integer
678 solution is $n_0(\varepsilon, \delta')$. (ii) For $p > \varepsilon$, with $\pi = \mathbb{P}_Q(T = 1)$ and T independent

679 of the real data, $\mathbb{P}(B = 1) = \pi(1 - p)^{n_1} + (1 - \pi)(1 - p)^{n_0} \leq (1 - \varepsilon)^{n_1} \leq \delta'$,
680 and $\leq \delta$ as soon as $\pi \leq \pi^*$ (the expression is affine in π and decreasing in
681 p). (iii) Write $a = \ln \delta / \ln(1 - \varepsilon)$, $a' = \ln \delta' / \ln(1 - \varepsilon)$ and $r = a - a' =$
682 $\ln(\delta'/\delta)/(-\ln(1 - \varepsilon))$. Since $n_0(\delta) \geq a$ and $n_0(\delta') < a' + 1$, $c > r - 1$; c being
683 an integer, $c \geq \lfloor r \rfloor = c^*$. $\square$

684 *Remark A.4* (numbers and checks). At $\varepsilon = \delta = 5\%$: $\delta' = 7.5, 10, 15, 20\%$
685 give $c = 8, 14, 22, 27$ ($-14\%, -24\%, -37\%, -46\%$ of 59). The exact form
686 is maximal on all 40 grid points $((1 - \varepsilon)^{n_0(\delta')} \leq \delta' < (1 - \varepsilon)^{n_0(\delta')-1}$: yes);
687 MC (10^5 runs per point) matches the closed form within 0.29 percentage
688 points. *Empirical C2* (procedure *B*, T = the one-sided CP bound of the
689 simulator's miss rate $\leq \varepsilon$, 10^5 runs, $\varepsilon = 5\%$, $d_{\min} = 32$, $K = 1$): with the
690 paired twin of 78 static scenes (180 events) the simulator passes in 98.1% of
691 runs; realised credit 7.9/13.7/21.6/26.5 events and false certification at the
692 worst case $p = \varepsilon$ equal to 7.2%/9.7%/14.7%/19.1% for $\delta' = 7.5/10/15/20\%$
693 — the credit is bought exactly with the extra error $\delta' - \delta$. The off-the-shelf
694 synthetic benchmark (30 events of length ≥ 32) never passes (0.0%): too few
695 synthetic events to bound its own miss rate below 5%.

696 *Appendix A.4. C3: the twin lemma*

697 **Lemma A.5** (C3). *For any pairing (M, M') of a real event and its twin,*
698 *$p \leq q + D$ with $D = \mathbb{P}(M = 1, M' = 0)$. No assumption links the twin to*
699 *reality.*

700 *Proof.* $p = \mathbb{P}(M = 1, M' = 1) + \mathbb{P}(M = 1, M' = 0) \leq \mathbb{P}(M' = 1) + D =$
701 $q + D$. $\square$

702 **Corollary A.6** (a per-camera twin does not save events). *If $q \leq q_U$ holds*
703 *with confidence $1 - \delta/2$ and $D \leq \varepsilon - q_U$ is certified for the same camera by*
704 *a zero-discordance success run at level $\delta/2$, the number of paired real events*
705 *is $\lceil \ln(\delta/2) / \ln(1 - (\varepsilon - q_U)) \rceil = 72, 91, 122$ for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$,*
706 *all larger than $n_0 = 59$. The twin can only help when D is calibrated once,*
707 *on many scenes, and transferred (C4).*

708 *Remark A.7* (event unit). With a uniform phase, $x_i = \mathbb{P}_\varphi(M_i = 1, M'_i = 0) =$
709 $|R'_i \setminus R_i|/K$, where R_i, R'_i are the residue sets of the real and twin hit frames
710 of event i (twin catch decided by the majority of three sprite variants). The
711 *worst-phase* indicator $W_i = \mathbf{1}\{R'_i \setminus R_i \neq \emptyset\}$ satisfies $\mathbb{P}(M_i = 1, M'_i = 0 \mid$
712 $\text{event}) = x_i \leq W_i$. All certificates below are stated for Bernoulli indicators

713 Z_i ; they hold verbatim with $Z_i = W_i$ and then bound the (larger) worst-
 714 phase discordance rate. This integer form is the one used for the reported
 715 certificates.

716 *Appendix A.5. fleet bound: event-weighted certificate for the calibrated fleet*

717 **Lemma A.8** (mean–median). *For $N \geq 2$ and $0 < p < 1 - 1/N$,
 718 $F_{N,p}(\lfloor Np \rfloor) > 1/4$. Consequently, if $\delta \leq 1/4$ and $F_{N,p}(x) < \delta$, then
 719 $x \leq Np - 1$.*

720 *Proof.* Greenberg and Mohri [51] prove $\mathbb{P}(B \geq \mathbb{E}B) > 1/4$ for $B \sim \text{Bin}(N, r)$
 721 with $r > 1/N$. Apply it to $B = N - \text{Bin}(N, p) \sim \text{Bin}(N, 1 - p)$ with $1 - p >$
 722 $1/N$: $\mathbb{P}(\text{Bin}(N, p) \leq Np) > 1/4$, and $\{\text{Bin}(N, p) \leq Np\} = \{\text{Bin}(N, p) \leq$
 723 $\lfloor Np \rfloor\}$. For the consequence: $F_{N,p}$ is non-decreasing, so $F_{N,p}(x) < \delta \leq$
 724 $1/4 < F_{N,p}(\lfloor Np \rfloor)$ forces $x < \lfloor Np \rfloor$, i.e. $x \leq \lfloor Np \rfloor - 1 \leq Np - 1$. *Check:*
 725 $\inf F_{N,p}(\lfloor Np \rfloor)$ over $N = 2..2000$ and a fine grid of p is 0.2500, approached
 726 at $N = 2$, $p \rightarrow 1/2$ (the constant $1/4$ is tight). $\square$

727 **Theorem A.9** (fleet bound). *Let $Z_1, \dots, Z_N$ be independent, $Z_i \sim \text{Bern}(d_i)$
 728 with arbitrary $d_i \in [0, 1]$ (heterogeneous across scenes and events), $X =$
 729 $\sum_i Z_i$ and $\bar{d} = \frac{1}{N} \sum_i d_i$. If $\delta \leq 1/4$ and $\bar{d} < 1 - 1/N$, then*

$$\mathbb{P}(U(X, N) < \bar{d}) \leq \delta.$$

730 *With $d_i = D_{s(i)}$ (the discordance of the scene of event i), $\bar{d} = \bar{D}_w =$
 731 $\sum_s n_s D_s / N$, the event-weighted discordance of the calibrated cameras; com-
 732 bined with Lemma A.5, $\bar{p}_w \leq \bar{q}_w + U(X, N)$ with probability at least $1 - \delta$.*

733 *Proof.* If $\bar{d} = 0$ there is nothing to prove. Otherwise let $x^* = \max\{x :$
 734 $F_{N,\bar{d}}(x) < \delta\}$ (if the set is empty the failure probability is 0). By monotonicity
 735 of U , $\{U(X, N) < \bar{d}\} = \{X \leq x^*\}$. By Lemma A.8, $x^* \leq N\bar{d} - 1$. Hoeffding’s
 736 Theorem 4 [50]: for a sum S of independent Bernoulli variables with mean
 737 $N\bar{d}$ and $0 \leq b \leq N\bar{d} - 1$, $\mathbb{P}(S \leq b) \leq F_{N,\bar{d}}(b)$. With $b = x^*$, $\mathbb{P}(X \leq x^*) \leq$
 738 $F_{N,\bar{d}}(x^*) < \delta$. $\square$

739 *Remark A.10* (what it does not say; checks). The guarantee is conditional on
 740 the calibrated scenes and weighted by events: it does not bound any single
 741 camera and does not transfer to a new camera (if scenes are random, X is
 742 over-dispersed; e.g. $D_s \in \{0, 1\}$ gives $\mathbb{P}(X = 0) = (1 - \mu)^m \gg (1 - \mu)^N$).
 743 *Checks.* Six synthetic heterogeneity patterns on a fixed configuration of 44
 744 scenes with $N = 304$ events (a calibration set from an earlier stage of the

study, used only as a test bed for the theorem): exact Poisson–binomial failure ≤ 0.0471 , MC (10^5) ≤ 0.0473 . Measured discordance probabilities of the study (event level, main domain, five operating points, as measured and rescaled to mean 2% and 5%; 12 cases): exact failure ≤ 0.0152 , MC ≤ 0.0151 , all $\leq \delta = 0.05$.

Appendix A.6. population bound: transfer to a new camera

Proposition A.11 (population bound). *Let scenes $S_1, \dots, S_m$ be i.i.d. from a scene population, each with $n_s \geq 1$ paired events whose discordance indicators are independent $\text{Bern}(D_{S_s})$ given the scene, $\mu = \mathbb{E}[D_S]$, and $J = \#\{s : X_s \geq 1\}$ with X_s the number of discordant events in scene s . Then*

$$\mathbb{P}(U(J, m) < \mu) \leq \delta.$$

For a new camera drawn from the same population, $\mathbb{E}[p_{\text{new}}] \leq \mathbb{E}[q_{\text{new}}] + U(J, m)$ with probability $\geq 1 - \delta$ over the calibration, and $\mathbb{P}(D_{\text{new}} > t) \leq U(J, m)/t$ (Markov).

Proof. Let $B_s = \mathbf{1}\{X_s \geq 1\}$. Given the scene, $\mathbb{P}(B_s = 1 \mid S_s) = 1 - (1 - D_{S_s})^{n_s} \geq D_{S_s}$ because $(1 - D)^n \leq 1 - D$ for $n \geq 1$. Hence $\beta_s := \mathbb{P}(B_s = 1) \geq \mu$, and the B_s are independent across scenes. By a coupling ($B_s = \mathbf{1}\{V_s \leq \beta_s\} \geq \mathbf{1}\{V_s \leq \mu\}$ with i.i.d. uniforms V_s), J stochastically dominates $\text{Bin}(m, \mu)$. Since $U(\cdot, m)$ is non-decreasing, $\mathbb{P}(U(J, m) < \mu) \leq \mathbb{P}(U(\text{Bin}(m, \mu), m) < \mu) \leq \delta$ by Clopper–Pearson validity. The corollary follows from Lemma A.5 applied to the new camera and averaged over the population. $\square$

Remark A.12 (price and checks). The resolution is set by the number of scenes: with $J = 0$, $U \leq 10\%$ needs $m \geq 29$ and $U \leq 5\%$ needs $m \geq 59$. *Checks.* The tight case $D_S \in \{0, 1\}$, $n_s = 1$, $m = 78$ (then $J \sim \text{Bin}(m, \mu)$ exactly), $\mu \in \{1, 2, 5, 10, 20\}\%$, 10^5 runs: maximum failure 0.0465. Leave-one-scene-out on the 78 static scenes at the post-hoc challenge point ($d_{\min} = 16$, $K = 16$): $D_s > U(J_{-s}, m - 1)$ for 1 of 76 scenes (the proposition bounds the population mean, not each scene).

Appendix A.7. per-camera form: a per-camera (quantile) certificate and its data requirement

Proposition A.13 (per-camera form). *Under the assumptions of Proposition A.11 with $n_s \geq n_{\min}$ for all s , let $t \in (0, 1)$, $\pi_t = \mathbb{P}(D_S > t)$ and*

777 $h_t = 1 - (1 - t)^{n_{\min}}$. Then

$$\mathbb{P}(\pi_t > U(J, m)/h_t) \leq \delta.$$

778 Hence, with $\hat{\gamma} = U(J, m)/h_t$, a new camera satisfies $p_{\text{new}} \leq q_{\text{new}} + t$ except
779 on a set of cameras of population mass at most $\hat{\gamma}$, with confidence $1 - \delta$.

780 *Proof.* $\mathbb{P}(B_s = 1) \geq \mathbb{P}(D_S > t)(1 - (1 - t)^{n_s}) \geq \pi_t h_t$. The coupling argument
781 of Proposition A.11 gives $J \succeq_{\text{st}} \text{Bin}(m, \pi_t h_t)$, so $\mathbb{P}(U(J, m) < \pi_t h_t) \leq \delta$,
782 which is the claim. $\square$

783 *Remark A.14* (data requirement; checks). With $J = 0$, $\hat{\gamma} \leq \gamma$ requires $m \geq$
784 $m_{\min}$ with $1 - \delta^{1/m_{\min}} \leq \gamma$ ($m_{\min} = 59$ for $\gamma = 5\%$, 29 for $\gamma = 10\%$) and $n_{\min} \geq$
785 $\ln(1 - U(0, m)/\gamma)/\ln(1 - t)$ events in every scene; the total $m n_{\min}$ decreases
786 to the floor $\approx \ln(1/\delta)/(\gamma t)$ as m grows. *Check:* two-point populations $D_S \in$
787 $\{0, t'\}$, $t' > t$, $m = 78$, 10^5 runs: maximum failure 0.0354.

788 *Appendix A.8. C5a: a pessimistic-length twin is conservative under periodic*
789 *sampling*

790 **Proposition A.15** (C5a). *Let an event occupy frames $F = \{o, \dots, o + d - 1\}$*
791 *and let the detector outcome on each frame be fixed (a deterministic function*
792 *of the frame). For the truncated event $F_L = \{o, \dots, o + L - 1\}$, $L \leq d$,*
793 *with hit set $H_L = H \cap F_L$, the miss probability under a uniform phase,*
794 *$m(L) = 1 - |H_L \bmod K|/K$, is non-increasing in L . Consequently a twin built*
795 *with a pessimistic (shorter) length L has $q_L \geq q$ and discordance $D_L \leq D$,*
796 *so the C3 bound $q_L + D_L$ remains valid and the miss term is conservative.*

797 *Proof.* $L \leq L'$ implies $H_L \subseteq H_{L'}$, hence $H_L \bmod K \subseteq H_{L'} \bmod K$ and
798 $m(L) \geq m(L')$. A twin catch at phase φ for the truncated twin implies
799 a catch for the full twin, so $\{M = 1, M'_L = 0\} \subseteq \{M = 1, M' = 0\}$ and
800 $D_L \leq D$; Lemma A.5 holds for any pairing. $\square$

801 *Remark A.16* (check). On the twin of the 78 static scenes, shortening every
802 event by $x \in \{10, 25, 50\}\%$ gives non-decreasing q for every event and every
803 $K \in \{1, 4, 8, 16, 48\}$ (all monotone: yes); e.g. at $d_{\min} = 32$, $K = 16$, q rises
804 from 0.95% to 1.68% at $x = 50\%$.

References

- [1] J. A. Hanley, If nothing goes wrong, is everything all right?, JAMA 249 (13) (1983) 1743. doi:10.1001/jama.1983.03330370053031.
- [2] W. Q. Meeker, G. J. Hahn, L. A. Escobar, Statistical Intervals, Wiley, 2017. doi:10.1002/9781118594841.
- [3] Y. Yoon, M.-S. Yang, C.-H. Bai, J. Shim, Bayesian zero-failure reliability demonstration tests: Sample size estimation and its limitation, Journal of Mechanical Science and Technology 39 (7) (2025) 3651–3666. doi:10.1007/s12206-025-2409-1.
- [4] W. Ding, X. Bai, Q. Wang, F. Long, H. Li, Z. Wu, J. Liu, H. Yao, H. Yang, A truncated test scheme design method for success-failure in-orbit tests, Reliability Engineering & System Safety 243 (2024) 109782. doi:10.1016/j.ress.2023.109782.
- [5] B. Wang, P. Jiang, B. Guo, A Bayesian design method for monopropellant engine system reliability qualification test plan, Reliability Engineering & System Safety 248 (2024) 110173. doi:10.1016/j.ress.2024.110173.
- [6] Y. Tan, P. Jiang, Y. Xing, J. Qi, Type-I censored reliability qualification test design for Weibull products based on expert judgments, Reliability Engineering & System Safety 262 (2025) 111074. doi:10.1016/j.ress.2025.111074.
- [7] Q. Wen, F. Machida, Reliability modeling for three-version machine learning systems through Bayesian networks, Reliability Engineering & System Safety 260 (2025) 111016. doi:10.1016/j.ress.2025.111016.
- [8] C. Paterson, R. Hawkins, C. Picardi, Y. Jia, R. Calinescu, I. Habli, Safety assurance of machine learning for autonomous systems, Reliability Engineering & System Safety 264 (2025) 111311. doi:10.1016/j.ress.2025.111311.
- [9] R. Aghazadeh-Chakherlou, Q. Guo, S. Khastgir, P. Popov, X. Zhang, X. Zhao, A hierarchical imprecise probability approach to reliability assessment of large language models, Reliability Engineering & System Safety 272 (2026) 112615. doi:10.1016/j.ress.2026.112615.

- 837 [10] N. Kalra, S. M. Paddock, Driving to safety: How many miles of driving
838 would it take to demonstrate autonomous vehicle reliability?, *Transportation Research Part A: Policy and Practice* 94 (2016) 182–193.
839 doi:10.1016/j.tra.2016.09.010.
840
- 841 [11] H. Tao, P. Jia, X. Wang, L. Wang, Reliability analysis of subsea control
842 module based on dynamic Bayesian network and digital twin, *Reliability*
843 *Engineering & System Safety* 248 (2024) 110153. doi:10.1016/j.ress
844 .2024.110153.
- 845 [12] Y. Ye, Q. Yang, J. Zhang, S. Meng, J. Wang, A dynamic data driven
846 reliability prognosis method for structural digital twin and experimental
847 validation, *Reliability Engineering & System Safety* 240 (2023) 109543.
848 doi:10.1016/j.ress.2023.109543.
- 849 [13] Y. Wang, B. Yang, J. Ma, Bayesian prediction of aerospace system
850 mission reliability with hierarchical and multi-fidelity test data fusion,
851 *Reliability Engineering & System Safety* 267 (2026) 111844. doi:
852 10.1016/j.ress.2025.111844.
- 853 [14] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, F. Diermeyer, Survey on
854 scenario-based safety assessment of automated vehicles, *IEEE Access* 8
855 (2020) 87456–87477. doi:10.1109/ACCESS.2020.2993730.
- 856 [15] X. Zhao, R. Aghazadeh-Chakherlou, C.-H. Cheng, P. Popov, L. Strigini,
857 On the need for a statistical foundation in scenario-based testing of
858 autonomous vehicles, in: *2025 IEEE 28th International Conference on*
859 *Intelligent Transportation Systems (ITSC)*, 2025, pp. 3998–4005. doi:
860 10.1109/ITSC60802.2025.11423546.
- 861 [16] F. Reway, A. Hoffmann, D. Wachtel, W. Huber, A. Knoll, E. Ribeiro,
862 Test method for measuring the simulation-to-reality gap of camera-
863 based object detection algorithms for autonomous driving, in: *2020*
864 *IEEE Intelligent Vehicles Symposium (IV)*, 2020, pp. 1249–1256. doi:
865 10.1109/IV47402.2020.9304567.
- 866 [17] T. R. Dieter, A. Weinmann, S. Jäger, E. Brucherseifer, Quantifying the
867 simulation–reality gap for deep learning-based drone detection, *Electronics* 12 (10) (2023) 2197. doi:10.3390/electronics12102197.
868

- [18] A. Kadian, J. Truong, A. Gokaslan, A. Clegg, E. Wijmans, S. Lee, M. Savva, S. Chernova, D. Batra, Sim2Real predictivity: Does evaluation in simulation predict real-world performance?, *IEEE Robotics and Automation Letters* 5 (4) (2020) 6670–6677. doi:10.1109/LRA.2020.3013848.
- [19] A. Kopp-Schneider, S. Calderazzo, M. Wiesenfarth, Power gains by using external information in clinical trials are typically not possible when requiring strict type i error control, *Biometrical Journal* 62 (2) (2020) 361–374. doi:10.1002/bimj.201800395.
- [20] B. Littlewood, L. Strigini, Validation of ultrahigh dependability for software-based systems, *Communications of the ACM* 36 (11) (1993) 69–80. doi:10.1145/163359.163373.
- [21] H. F. Martz, R. A. Waller, A Bayesian zero-failure (baze) reliability demonstration testing procedure, *Journal of Quality Technology* 11 (3) (1979) 128–138. doi:10.1080/00224065.1979.11980894.
- [22] K. J. Wilson, M. Farrow, Assurance for sample size determination in reliability demonstration testing, *Technometrics* 63 (4) (2021) 523–535. doi:10.1080/00401706.2020.1867646.
- [23] H. Zheng, J. Yang, Y. Zhao, Reliability demonstration test plan for degraded products subject to Gamma process with unit heterogeneity, *Reliability Engineering & System Safety* 240 (2023) 109617. doi:10.1016/j.ress.2023.109617.
- [24] X. Zhao, B. Littlewood, A. Povyakalo, L. Strigini, D. Wright, Conservative claims for the probability of perfection of a software-based system using operational experience of previous similar systems, *Reliability Engineering & System Safety* 175 (2018) 265–282. doi:10.1016/j.ress.2018.03.032.
- [25] S. Li, J. Li, J. Liu, C. Xu, Z. Dong, Scenario generation for testing autonomous escort operations of intelligent tugs based on surrogate modeling and hybrid sampling, *Reliability Engineering & System Safety* 268 (2026) 111975. doi:10.1016/j.ress.2025.111975.

- 900 [26] S. Riedmaier, B. Danquah, B. Schick, F. Diermeyer, Unified framework
901 and survey for model verification, validation and uncertainty quantifica-
902 tion, *Archives of Computational Methods in Engineering* 28 (4) (2021)
903 2655–2688. doi:10.1007/s11831-020-09473-7.
- 904 [27] M. H. Amini, S. Nejati, Bridging the gap between real-world and syn-
905 thetic images for testing autonomous driving systems, in: *Proceedings of*
906 *the 39th IEEE/ACM International Conference on Automated Software*
907 *Engineering*, 2024, pp. 732–744. doi:10.1145/3691620.3695067.
- 908 [28] A. N. Angelopoulos, S. Bates, C. Fannjiang, M. I. Jordan, T. Zrnic,
909 Prediction-powered inference, *Science* 382 (6671) (2023) 669–674. doi:
910 10.1126/science.adi6000.
- 911 [29] A. Fisch, J. Maynez, R. Hofer, B. Dhingra, A. Globerson, W. Cohen,
912 Stratified prediction-powered inference for effective hybrid evaluation of
913 language models, in: *Advances in Neural Information Processing Sys-*
914 *tems* 37, 2024, pp. 111489–111514. doi:10.52202/079017-3541.
- 915 [30] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn
916 then test: Calibrating predictive algorithms to achieve risk control, *The*
917 *Annals of Applied Statistics* 19 (2) (2025). doi:10.1214/24-A0AS1998.
- 918 [31] F. de Grancey, J.-L. Adam, L. Alecu, S. Gerchinovitz, F. Mamalet,
919 D. Vigouroux, Object detection with probabilistic guarantees: A con-
920 formal prediction approach, in: *Lecture Notes in Computer Science*,
921 2022, pp. 316–329. doi:10.1007/978-3-031-14862-0_23.
- 922 [32] Z. Mei, A. Dixit, M. Booker, E. Zhou, M. Storey-Matsutani, A. Z.
923 Ren, O. Shorinwa, A. Majumdar, Perceive with confidence: Statis-
924 tical safety assurances for navigation with learning-based perception,
925 *The International Journal of Robotics Research* 45 (6) (2026) 938–967.
926 doi:10.1177/02783649251378151.
- 927 [33] S. Yuan, J. Li, C. Wang, X. Zhang, Conformal machine learning for reli-
928 able anomaly detection in industrial cyber-physical systems, *Reliability*
929 *Engineering & System Safety* 274 (2026) 112417. doi:10.1016/j.res
930 .2026.112417.
- 931 [34] H. Zheng, J. Yang, H. Xu, Y. Zhao, Reliability acceptance sampling plan
932 for degraded products subject to Wiener process with unit heterogeneity,

- 933 Reliability Engineering & System Safety 229 (2023) 108877. doi:10.1
934 016/j.ress.2022.108877.
- 935 [35] X. Zhao, V. Robu, D. Flynn, K. Salako, L. Strigini, Assessing the safety
936 and reliability of autonomous vehicles from road testing, in: 2019 IEEE
937 30th International Symposium on Software Reliability Engineering (IS-
938 SRE), 2019, pp. 13–23. doi:10.1109/ISSRE.2019.00012.
- 939 [36] X. Zhao, K. Salako, L. Strigini, V. Robu, D. Flynn, Assessing safety-
940 critical systems from operational testing: A study on autonomous ve-
941 hicles, Information and Software Technology 128 (2020) 106393. doi:
942 10.1016/j.infsof.2020.106393.
- 943 [37] B. Littlewood, K. Salako, L. Strigini, X. Zhao, On reliability assessment
944 when a software-based system is replaced by a thought-to-be-better one,
945 Reliability Engineering & System Safety 197 (2020) 106752. doi:10.1
946 016/j.ress.2019.106752.
- 947 [38] S. Morita, P. F. Thall, P. Müller, Determining the effective sample size
948 of a parametric prior, Biometrics 64 (2) (2008) 595–602. doi:10.1111/
949 j.1541-0420.2007.00888.x.
- 950 [39] M.-H. Chen, J. G. Ibrahim, Power prior distributions for regression mod-
951 els, Statistical Science 15 (1) (2000). doi:10.1214/ss/1009212673.
- 952 [40] B. P. Hobbs, B. P. Carlin, S. J. Mandrekar, D. J. Sargent, Hierarchical
953 commensurate and power prior models for adaptive incorporation of
954 historical information in clinical trials, Biometrics 67 (3) (2011) 1047–
955 1056. doi:10.1111/j.1541-0420.2011.01564.x.
- 956 [41] P. Bishop, A. Povyakalo, Deriving a frequentist conservative confidence
957 bound for probability of failure per demand for systems with different
958 operational and test profiles, Reliability Engineering & System Safety
959 158 (2017) 246–253. doi:10.1016/j.ress.2016.08.019.
- 960 [42] R. Dwight, W. Li, F. van Rooij, P. Scarf, Maintenance planning using
961 a digital twin: principles and case studies, Reliability Engineering &
962 System Safety 265 (2026) 111496. doi:10.1016/j.ress.2025.111496.
- 963 [43] K. Saad, A. Yousif, C. F. Elias, A digital twin approach for perception
964 development and validation in autonomous vehicles*, in: 2025 IEEE

- Intelligent Vehicles Symposium (IV), 2025, pp. 2577–2584. doi:10.1109/IV64158.2025.11097489.
- [44] M. Barbier, A. Renzaglia, J. Quilbeuf, L. Rummelhard, A. Paigwar, C. Laugier, A. Legay, J. Ibanez-Guzman, O. Simonin, Validation of perception and decision-making systems for autonomous driving via statistical model checking, in: 2019 IEEE Intelligent Vehicles Symposium (IV), 2019, pp. 252–259. doi:10.1109/IVS.2019.8813793.
- [45] R. Luo, H. Yang, M. Watson, A. Sharma, S. Veer, E. Schmerling, M. Pavone, Sim2Val: Leveraging correlation across test platforms for variance-reduced metric estimation, Conference on Robot Learning (CoRL) 2025; cited as arXiv preprint arXiv:2506.20553 (2025). doi:10.48550/arXiv.2506.20553.
- [46] D. Zhu, S. Oh, R. Huang, Z. Huang, J. Ma, C. Tang, SCAPE: Scenario-conditioned simulation-augmented policy evaluation, Preprint arXiv:2608.19425 (not peer reviewed) (2026). doi:10.48550/arXiv.2608.19425.
- [47] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends® in Machine Learning 16 (4) (2023) 494–591. doi:10.1561/2200000101.
- [48] H. Yang, M. Pavone, Object pose estimation with statistical guarantees: Conformal keypoint detection and geometric uncertainty propagation, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 8947–8958. doi:10.1109/CVPR52729.2023.00864.
- [49] A. Vacavant, T. Chateau, A. Wilhelm, L. Lequière, A benchmark dataset for outdoor foreground/background extraction, in: Lecture Notes in Computer Science, 2013, pp. 291–300. doi:10.1007/978-3-642-37410-4_25.
- [50] W. Hoeffding, On the distribution of the number of successes in independent trials, The Annals of Mathematical Statistics 27 (3) (1956) 713–721. doi:10.1214/aoms/1177728178.

- 996 [51] S. Greenberg, M. Mohri, Tight lower bound on the probability of a
 997 binomial exceeding its expectation, *Statistics & Probability Letters* 86
 998 (2014) 91–98. doi:10.1016/j.spl.2013.12.009.
- 999 [52] Y. Wang, P.-M. Jodoin, F. Porikli, J. Konrad, Y. Benezeth, P. Ishwar,
 1000 CDnet 2014: An expanded change detection benchmark dataset, in:
 1001 2014 IEEE Conference on Computer Vision and Pattern Recognition
 1002 Workshops, 2014, pp. 393–400. doi:10.1109/CVPRW.2014.126.
- 1003 [53] C. Cuevas, E. M. Yáñez, N. García, Labeled dataset for integral eval-
 1004 uation of moving object detection algorithms: LASIESTA, *Computer*
 1005 *Vision and Image Understanding* 152 (2016) 103–117. doi:10.1016/j.
 1006 cviu.2016.08.005.
- 1007 [54] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once:
 1008 Unified, real-time object detection, in: 2016 IEEE Conference on Com-
 1009 puter Vision and Pattern Recognition (CVPR), 2016, pp. 779–788.
 1010 doi:10.1109/CVPR.2016.91.
- 1011 [55] Ultralytics, Ultralytics YOLO (yolo26s-seg), [https://github.com/ult](https://github.com/ultralytics/ultralytics)
 1012 [ralytics/ultralytics](https://github.com/ultralytics/ultralytics), accessed 2026-09-25.
- 1013 [56] Y. Cabon, N. Murray, M. Humenberger, Virtual KITTI 2,
 1014 arXiv:2001.10773 (2020). doi:10.48550/arXiv.2001.10773.
- 1015 [57] MPFB 2: MakeHuman plugin for Blender, version 2.0.17, [https://ex](https://extensions.blender.org/add-ons/mpfb/)
 1016 [tensions.blender.org/add-ons/mpfb/](https://extensions.blender.org/add-ons/mpfb/), bundled assets CC0; accessed
 1017 2026-09-27.
- 1018 [58] H. Shimodaira, Improving predictive inference under covariate shift by
 1019 weighting the log-likelihood function, *Journal of Statistical Planning and*
 1020 *Inference* 90 (2) (2000) 227–244. doi:10.1016/S0378-3758(00)00115
 1021 -4.
- 1022 [59] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits
 1023 illustrated in the case of the binomial, *Biometrika* 26 (4) (1934) 404–
 1024 413. doi:10.1093/biomet/26.4.404.